Donghyeon Soon

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Education

DGIST (Daegu Gyeongbuk Institute of Science & Technology) *Daegu, South Korea*
B.S. in Computer & Electronics Engineering *Feb 2022 – Aug 2026 (expected)*

- Focus: Computer Science, Artificial Intelligence, 3D Vision, Robotics
- Coursework: Computer Vision, Machine Learning, Deep Learning, Graphics

Research Interests

- 3D vision and generative modeling for reconstructing, animating, and generating dynamic 3D/4D content.
- Human and animal reconstruction using synthetic data pipelines, parametric models, and image/video-to-3D/4D methods.

Experience

Research Intern, DGIST *Dec 2025 – Present*
Intelligent Systems and Learning Lab (Prof. Daehee Park): scene-aware human motion prediction.

KIC Tech Frontier Program Participant *Jun 2025 – Aug 2025*
UC Berkeley & KIC: real-time 3D human motion estimation and technology commercialization.

Research Intern, DGIST *Jan 2025 – Feb 2025*
Computer Vision Lab (Prof. Sunghoon Im): 3D Gaussian editing and style transfer.

Research Intern, UNIST *Jun 2023 – Dec 2024*
3D Vision & Robotics Lab (Prof. Kyungdon Joo): animatable 3D reconstruction; DogRecon (IJCV 2025).

Publications & Manuscripts

Preprints / Under Review

Feed-Forward Animatable Animal Reconstruction Under review

Multimodal Clinical Prediction Research Under review

Expressive Multi-Dancer Dataset Under review

International

SGIP: Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction 2026
Donghyeon Soon*, Jiil Yoon*, Hyeondong Kim, Daehee Park†
Advances in Neural Information Processing Systems (NeurIPS), 2026
Paper / Project page / Code

AniGauss: Toward Animatable Animal Reconstruction from Single In-the-Wild Images via Topology-Aware Gaussians 2026
Gyeongsu Cho, Changwoo Kang, **Donghyeon Soon**, Hezhen Hu, Kyungdon Joo†
CV4Animals Workshop at CVPR 2026
Paper / Project page

International (continued)

WildAni4D: Towards 4D Animal Mesh Reconstruction 2026
Gyeongsu Cho, Hezhen Hu, **Donghyeon Soon**, Changwoo Kang, Kyungdon Joo†
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026 Findings

Not Like Transformers: Drop the Beat Representation for Dance Generation with Mamba-Based Diffusion Model 2026
Sangjune Park, Inhyeok Choi, **Donghyeon Soon**, Youngwoo Jeon, Kyungdon Joo†
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026
Also presented at ICCV 2025 Workshop on Generative Models for Audio-Visual Creation (Gen4AVC)
Paper / Project page

DogRecon: Canine Prior-Guided Animatable 3D Gaussian Dog Reconstruction From a Single Image 2025
Gyeongsu Cho, Changwoo Kang, **Donghyeon Soon**, Kyungdon Joo†
International Journal of Computer Vision (IJCV), 2025
Also presented at CVPR 2024 Workshop on Computer Vision for Animals (CV4Animals)
Paper / Project page

Domestic

Consistent Image Generation via Aligned-Mean Flow Coupling 2027
Minseok Oh, **Donghyeon Soon**, Jeong Hoon Choi, Jeong Tae Young
Korea Artificial Intelligence Conference, 2027

Enhancing Foothold Planning for Locomotion on Curved Discrete Terrain 2026
Taebok Lee*, **Donghyeon Soon***, Kyungdon Joo†
ICROS, 2026 (Undergraduate Paper Award)

Improving Geometric Processing with Plane-aware Perturbation Uncertainty and Inference-time Control for Monocular Depth Estimation 2026
Donghyeon Soon*, Minseok Oh*, Sangchul Lee†
Korea Artificial Intelligence Conference, 2026

CC-RAG: Confidence-Controlled Retrieval-Augmented Generation 2026
Donghyeon Soon*, Minseok Oh*, Sangchul Lee†
Korea Artificial Intelligence Conference, 2026

Next-Scale Inversion Based Reference-Guided T2I Style Transfer 2026
Minseok Oh*, **Donghyeon Soon***, Sangchul Lee†
Korea Artificial Intelligence Conference, 2026 (Outstanding Paper Award)

Grid-based Local Homography Model for Panorama Stitching under Parallax 2026
Donghyeon Soon, Minseok Oh, Taebok Lee
Korea Artificial Intelligence Conference, 2026

Ghost Flare Localization and Removal Via Convex Sparse Template Selection 2026
Minseok Oh*, **Donghyeon Soon***, Sangchul Lee†
Korea Artificial Intelligence Conference, 2026

Enhancing Underwater Object Detection via Conditional Diffusion Model based Preprocessing and Augmentation 2025
Donghyeon Soon*, Taebok Lee*
Korea Artificial Intelligence Conference, 2025 (Outstanding Paper Award)
Paper

** Equal contribution. † Advisor/corresponding author.*

Selected Projects

Scene-aware Human Motion Prediction *Dec 2025 – Present*
DGIST ISL Lab

- Developed a framework for human motion prediction that captures interactions with the surrounding environment.
- Targets improved future trajectory and pose estimation in scene-aware settings.

Expressive Multi-Dancer Dataset *2026*
UT Austin & UNIST

- Built a large-scale expressive multi-dancer dataset with automated annotations.
- Includes world-grounded multi-person SMPL-X motion, detailed hand articulation, facial expressions, audio, and structured text descriptions.

AniGauss: Topology-Aware Animatable Animal Reconstruction *2026*
UNIST & UT Austin

- Designed a feed-forward single-image animal reconstruction pipeline using canonical view generation and SMAL-topology-aware Gaussians.
- Project page: vision3d-lab.github.io/anigauss

KIC Tech Frontier Program: Real-Time 3D Human Motion Estimation *Jun 2025 – Aug 2025*
UC Berkeley & KIC

- Conducted a startup-oriented project on real-time 3D human motion estimation.
- Participated in entrepreneurship and technology commercialization activities through the KIC Tech Frontier Program.

Skills

Programming: Python, C++, C, CUDA, Bash

Frameworks & Tools: PyTorch, OpenCV, Git, LaTeX, Gaussian Splatting pipelines

Domains: 3D/4D Vision, Generative Modeling, Diffusion Models, Gaussian Splatting, Human Motion Prediction, Human/Animal Reconstruction, Parametric Models (SMAL, SMPL, SMPL-X), Pose Estimation, Synthetic Data Generation, Robotics